Telecom Customer Churn Analysis & Prediction

From Data to Decision: Solving a Real-World Business Problem

7,043
Customers

21
Features

26.5%
Churn Rate

~93%
Model AUC

Presentation Agenda

What we will cover today

01 **Business Problem**

Why churn matters and what we set out to solve

02 **Data & Cleaning**

Dataset overview and wrangling steps

03 **Exploratory Analysis**

Key patterns and churn drivers discovered

04 **Machine Learning Model**

Approach, results, and feature importance

05 **Insights & Dashboard**

Business findings and Tableau visualisation

06 **Recommendations**

5 actionable strategies to reduce churn

The Business Problem

Why customer churn is a critical business challenge

The Challenge

A telecom company is losing 1 in 4 customers to churn. Every churned customer means lost recurring revenue and increased cost to acquire a replacement.

Our Business Question

"What factors drive customer churn, and can we predict who will leave next?"

5x Cost to acquire vs retain a customer

26.5% Of customers have already left

$13M+ Estimated by multiplying the average monthly spend of churned users by 12 months.

Data Wrangling & Cleaning

IBM Telco Customer Churn Dataset: Kaggle

7,043
Customers

21
Columns

11
Blank values cleaned

1
New feature engineered

TotalCharges → Float

Stored as string with 11 blank values. Converted via pd.to_numeric(), blanks filled with 0 (new customers, tenure=0)

SeniorCitizen → Yes/No

Was stored as 0/1 integer. Mapped to readable Yes/No labels for consistency and Tableau compatibility

Dropped customerID

Unique identifier with no predictive value. Removed to keep the dataset lean

Encoded Churn to Binary

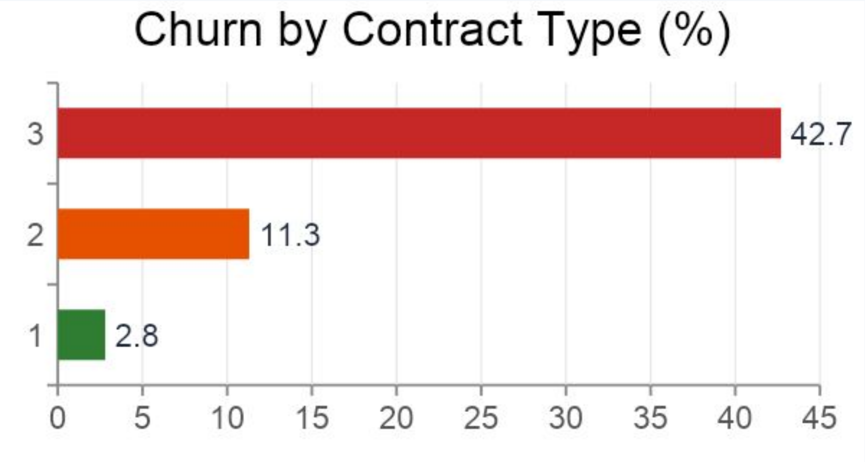
Converted Yes/No to 1/0 for correlation analysis and machine learning modelling

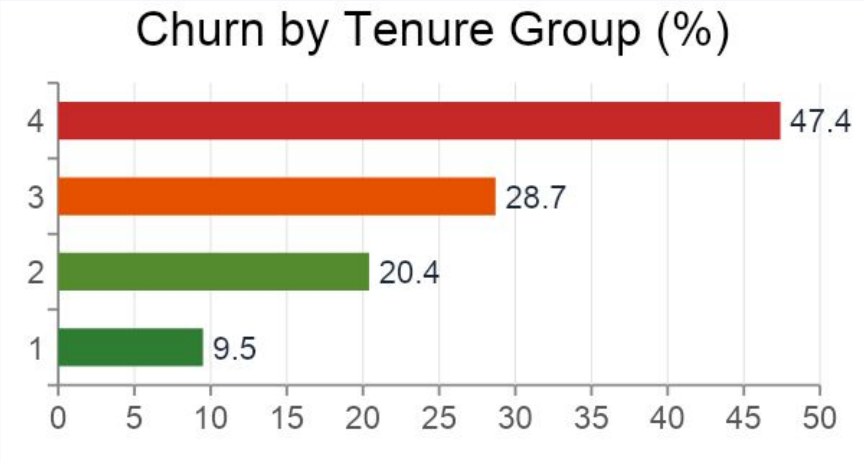
Engineered tenure_group

Created 4 tenure buckets: 0-12, 13-24, 25-48, 49-72 months for richer EDA and Tableau visuals

Exploratory Data Analysis

10 visualisations produced: here are the 4 most important findings





Contract is the #1 lever

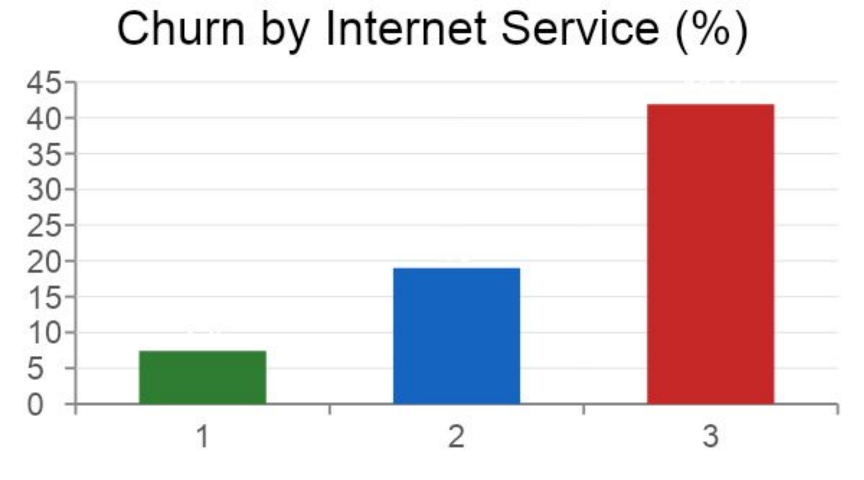
Month-to-month customers churn at 42.7%, 15× more than two-year contract holders at 2.8%.

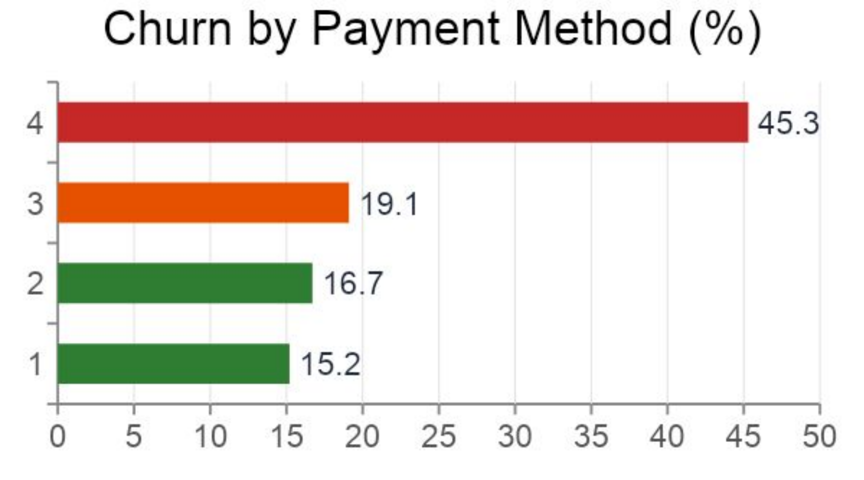
Early tenure = highest risk

Customers in their first year are most vulnerable at 47.4% churn rate. Loyalty grows with time.

More Key Findings

Internet service type and payment method reveal further churn patterns





Fiber optic paradox

The premium service has the highest churn (41.9%).
Customers pay the most but feel the least value.

Electronic check = risk flag

45.3% churn rate — 3× higher than auto-pay users.
Switching to auto-pay is a quick retention win.

Machine Learning Model

Training two models and selecting the best performer

Raw Data → Encode Categoricals → SMOTE Balancing → 80/20 Train-Test → Scale Features → Train Models

Logistic Regression

Baseline

Accuracy	~80%
ROC-AUC	~0.88
Precision	~79%
Recall	~80%

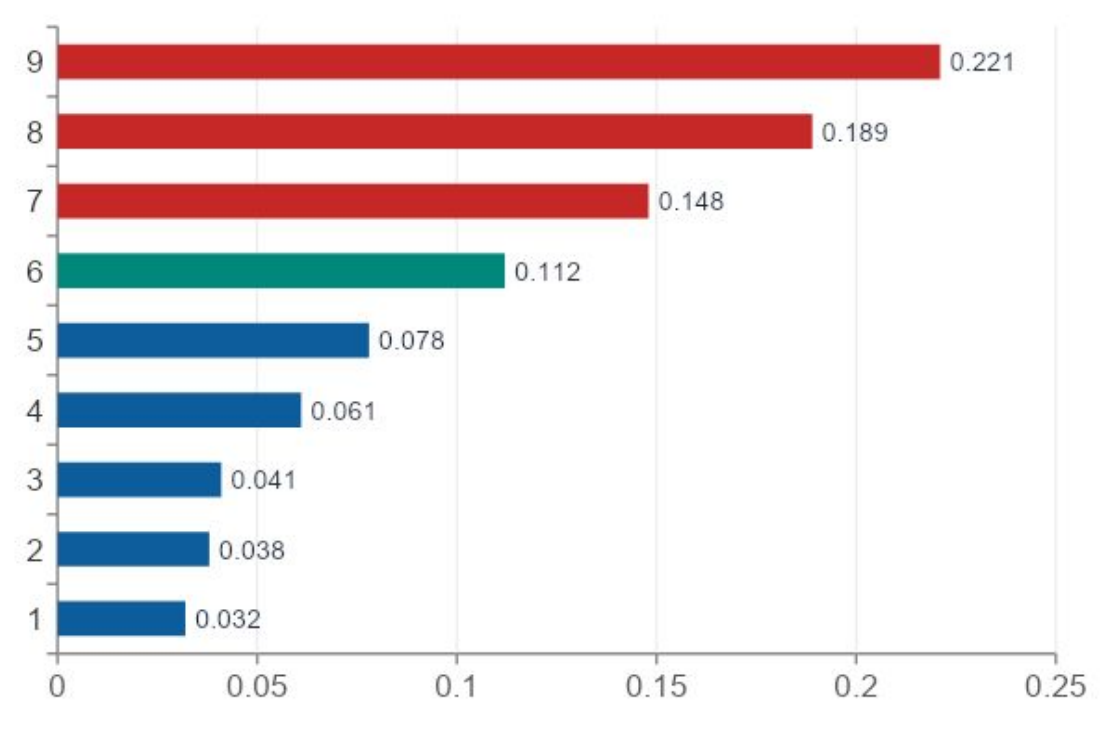
Random Forest ✓

Selected

Accuracy	~85%
ROC-AUC	~0.93
Precision	~85%
Recall	~85%

What Predicts Churn?

Top features identified by the Random Forest model



01 **Tenure**
Longest relationship = lowest risk

02 **TotalCharges**
High spend = high stakes in leaving

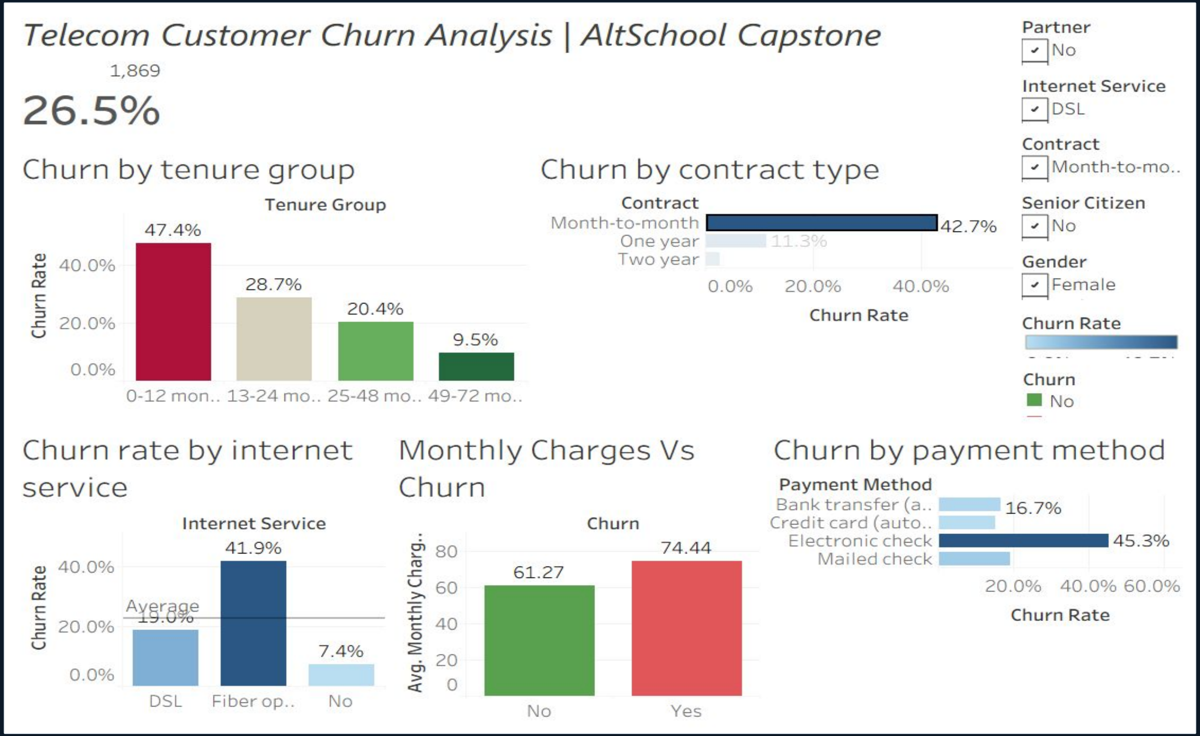
03 **MonthlyCharges**
Price sensitivity signals churn

04 **Contract Type**
Most actionable retention lever

05 **InternetService**
Fiber optic users at high risk

Tableau Dashboard

Interactive business intelligence dashboard built on the cleaned dataset



Interactive filters: Gender · Senior Citizen · Contract · Internet · Partner

Dashboard Components

1. KPI: 26.5% churn rate & 1,869 customers
2. Churn by Contract Type
3. Churn by Tenure Group
4. Churn by Internet Service
5. Monthly Charges vs Churn
6. Churn by Payment Method

Key Insights

9 data-backed findings from the analysis

🔴 **Month-to-month = 42.7% churn**

15× higher than two-year customers

🔴 **New customers (0-12 mo) = 47.4%**

Highest risk window is the first year

🔴 **Fiber optic = 41.9% churn**

Premium users feel the least value

🟡 **Electronic check = 45.3% churn**

3× more likely to leave vs auto-pay

🟡 **Senior citizens churn at ~41%**

vs 24% for non-seniors

🟡 **Higher charges = higher churn**

$74.44 avg for churned vs $61.27 retained

🟢 **49-72 month customers = 9.5% churn**

Long tenure is the strongest loyalty signal

🟢 **Auto-pay users rarely churn**

15-17% churn vs 45% for electronic check

🟢 **Two-year contracts = 2.8% churn**

Commitment creates stability

5 Business Recommendations

Evidence-backed strategies to reduce churn from 26.5% to below 20%

01 Incentivise longer contracts

Offer 10-20% discount to customers who upgrade from monthly to annual

Impact: ↓ ~15% churn

02 Improve early onboarding

90-day program: check-ins, offers, dedicated new-customer support line

Impact: Target 47% risk group

03 Fix Fiber Optic value gap

Customer satisfaction survey + year-one price-lock guarantee

Impact: Highest revenue impact

04 Promote auto-pay enrolment

Offer $5/mo discount to switch from electronic check to auto-pay

Impact: 45% → 16% churn

05 Deploy churn prediction model

Flag >70% risk customers in CRM, trigger retention offer 30 days ahead

Impact: Proactive vs reactive

Conclusion

Phase	What We Did	Outcome
Problem Framing	Defined business question around customer churn	Clear, measurable objective
Data Wrangling	Fixed TotalCharges, encoded SeniorCitizen, engineered tenure_group	Clean, analysis-ready dataset
EDA (10 charts)	Explored all key churn drivers visually	9 data-backed insights
ML Modelling	Logistic Regression + Random Forest with SMOTE + CV	~85% accuracy, ~0.93 AUC
Recommendations	5 evidence-based business strategies	Actionable, impact-estimated
Tableau Dashboard	6-chart interactive dashboard with filters	Stakeholder-ready story

Yes, we can predict who can leave next

Thank You

Questions & Discussion